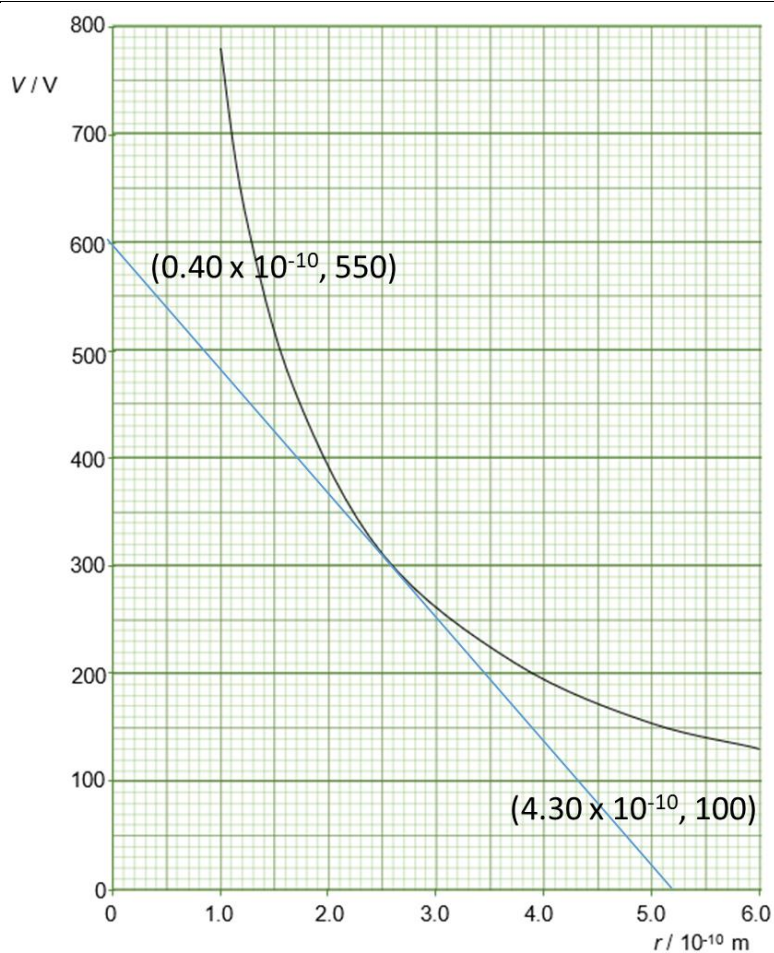


Qn 4	Suggested Solutions	Mark
4(a)	The electric potential at a point is the work done per unit positive charge , in bringing a small test charge from infinity to that point.	B1
4(b)(i)	At $r = 1.0 \times 10^{-10}$ m, the potential is 780 V. $V = \frac{Q}{4\pi\epsilon_0 r}$ $= \frac{ne}{4\pi\epsilon_0 r}, \text{ where } n \text{ is number of protons}$ $n = \frac{V(4\pi\epsilon_0 r)}{e} = \frac{780(4\pi)(8.85 \times 10^{-12})(1.0 \times 10^{-10})}{1.60 \times 10^{-19}}$ $= 54$	M1 M1 M1
4(b)(ii)	The distance 2.0×10^{-8} m is considerably greater than the diameters or radii of the nucleus and the proton.	B1
(b)(iii) 1.	The magnitude of the electric field strength at a point is equal to the gradient of the potential-distance curve at that point.	B1
2.	<u>Approach 1</u> determine E by finding gradient of tangent at $r = 2.6 \times 10^{-10}$ m $E = -\frac{\Delta V}{\Delta r} = -\frac{550 - 100}{(0.40 - 4.30) \times 10^{-10}} = 1.15 \times 10^{12} \text{ NC}^{-1}$ $F_{\text{electric}} = qE$ $= (1.60 \times 10^{-19})(1.1538 \times 10^{12})$ $= 1.85 \times 10^{-7} \text{ N}$ Direction: radially away from the nucleus gradient – 1 mark substitution of values into force equation – 1 mark value of force – 1 mark direction (radially must be present) – 1 mark	M1 M1 A1 A1

Approach 2

From Fig. 4.1, when $r = 2.6 \times 10^{-10} \text{ m}$, $V = 300 \text{ V}$

$$\begin{aligned}
 F_{\text{electric}} &= qE = q \left(\frac{dV}{dr} \right) \\
 &= q \left(\frac{V}{r} \right) = e \left(\frac{54e}{4\pi\epsilon_0 r} \right) \left(\frac{1}{r} \right) \\
 &= (1.60 \times 10^{-19}) \left(\frac{300}{2.6 \times 10^{-10}} \right) \\
 &= 1.85 \times 10^{-7} \text{ N}
 \end{aligned}$$

radially away from the nucleus

M1

M1

A1

A1

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C2

Q5	Suggested Solutions	Mark
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